
Algorithm: Eye Feature Extraction & Classification

Input: $\mathcal{D}_{\text{img}}^L, \mathcal{D}_{\text{img}}^R, \mathcal{D}_{\text{pat}}, T_{\text{thres}}$ **Output:** $\mathbf{A}_{\text{metrics}}, \mathbf{R}_{\text{class}}$ **Initialize:** $\mathbf{F}_{\text{eye}} \leftarrow \emptyset, \mathbf{Y} \leftarrow \{1\}^{N_{\text{norm}}} \oplus \{0\}^{N_{\text{pat}}}, \mathbf{B}_{\text{train}} \leftarrow \{\mathbf{X}_{\text{norm}}, \mathbf{X}_{\text{pat}}\}, N_{\text{tree}} \leftarrow 90$

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1 Function Image processing ( $\mathcal{D}_{\text{img}}^L, \mathcal{D}_{\text{img}}^R, T_{\text{thres}}$ )
2   for  $\mathbf{I}_k \in \mathcal{D}_{\text{img}}^L \cup \mathcal{D}_{\text{img}}^R$  do
3      $(\hat{x}_c, \hat{y}_c, \hat{w}) \leftarrow \text{HaarCascade}(\mathbf{I}_k)$                                 ▷ Detect eye region
4      $\mathcal{E}_k \leftarrow \text{FitEllipse}(\mathbf{I}_k, \hat{x}_c, \hat{y}_c; T_{\text{thres}})$                     ▷ Fit pupil ellipse
5      $(\mathcal{C}_k^{\text{in}}, \mathcal{C}_k^{\text{out}}) \leftarrow \text{EdgeConv}(\mathbf{I}_k, \mathcal{E}_k)$                     ▷ Detect eye corners
6      $\Delta x, \Delta y, \sigma_{\text{speed}}, \mu_{\text{disp}} \leftarrow \text{ComputeFeatures}(\mathbf{I}_k, \mathcal{C}_k^{\text{in}}, \mathcal{C}_k^{\text{out}})$     ▷ Compute features
7      $\mathbf{F}_{\text{eye}} \leftarrow \mathbf{F}_{\text{eye}} \oplus [\Delta x, \Delta y, \sigma_{\text{speed}}, \mu_{\text{disp}}]$     ▷ Aggregate features
8   end
9   return  $\mathbf{F}_{\text{eye}}$ 
10 end
11 Function Classification and validation ( $\mathbf{F}_{\text{eye}}, \mathbf{Y}, N_{\text{tree}}$ )
12    $\mathbf{D}_{\text{all}} \leftarrow \text{Concatenate}(\mathbf{F}_{\text{eye}}, \mathbf{Y})$                                 ▷ Classify and validate
13   for  $i \in 1 \dots (N_{\text{norm}} + N_{\text{pat}})$  do
14      $\mathbf{D}_{\text{train}} \leftarrow \mathbf{D}_{\text{all}} \setminus \mathbf{D}_{\text{all}}(i)$     $\mathbf{D}_{\text{test}} \leftarrow \mathbf{D}_{\text{all}}(i)$         ▷ Split data
15      $\pi_{\theta} \leftarrow \text{RandomForest}(\mathbf{D}_{\text{train}}, N_{\text{tree}})$     ▷ Train random forest
16      $\hat{y}_i \leftarrow \pi_{\theta}(\mathbf{D}_{\text{test}}(\text{features}))$         ▷ Make prediction
17      $\mathbf{R}_{\text{class}}(i) \leftarrow \mathbb{I}(\hat{y}_i == \mathbf{Y}(i))$         ▷ Check accuracy
18   end
19   return  $\mathbf{R}_{\text{class}}$ 
20 end
21 Function Calculate average accuracy ( $\mathbf{R}_{\text{class}}$ )
22    $\mathbf{A}_{\text{metrics}} \leftarrow \text{Mean}(\mathbf{R}_{\text{class}})$                                 ▷ Calc average accuracy
23    $\mathbf{R}_{\text{class}} \leftarrow \text{Reshape}(\mathbf{R}_{\text{class}})$                 ▷ Reshape results
24   return  $\mathbf{A}_{\text{metrics}}, \mathbf{R}_{\text{class}}$ 
25 end
26 Function main ()
27    $\mathbf{F}_{\text{eye}} \leftarrow \text{Image processing}(\mathcal{D}_{\text{img}}^L, \mathcal{D}_{\text{img}}^R, T_{\text{thres}})$ 
28    $\mathbf{R}_{\text{class}} \leftarrow \text{Classification and validation}(\mathbf{F}_{\text{eye}}, \mathbf{Y}, N_{\text{tree}})$ 
29    $\mathbf{A}_{\text{metrics}},$ 
30    $\mathbf{R}_{\text{class}} \leftarrow \text{Calculate average accuracy}(\mathbf{R}_{\text{class}})$ 
31   Output  $\mathbf{A}_{\text{metrics}}, \mathbf{R}_{\text{class}}$ 
32 end
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